

LEARNING & SKILL DEVELOPMENT REPORT

AquaSense Phase 1

Multidisciplinary Learning Journey

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Prepared as a record of technical learning, application, problem-solving, and professional development.

1. Introduction

As part of the AquaSense Phase 1 development journey, we were guided through a multidisciplinary learning process that extended beyond the development of a single hardware project. The primary objective was to understand how different engineering and technology tools can be combined to transform an idea into a practical and presentable solution.

Under the guidance of Mr. Mahesh Devagiri, we explored website development, application development, 3D modelling using Autodesk Fusion 360, custom PCB design using KiCad, and critical thinking and problem-solving. The learning process helped us understand how to analyze a problem, evaluate possible solutions, select appropriate technologies, and develop a complete system systematically.

2. Objectives of the Learning

- To understand how an idea can be converted into a functional digital and physical solution.
- To learn the fundamentals of website and application development.
- To understand user-interface planning and user-oriented solution design.
- To learn 3D modelling and product design using Autodesk Fusion 360.
- To understand custom PCB design using KiCad.
- To develop critical thinking and engineering decision-making.
- To understand how software, electronics, mechanical design, and documentation work together.
- To develop a structured approach toward innovation and prototyping.

3. Learning Area 1 – Website Development

We learned that a website is not simply a collection of pages; it is a platform through which a project can be explained, documented, demonstrated, and presented professionally.

Key areas learned

- Basic website structure and organization.
- Creating sections for project information.
- Organizing information logically.
- Designing attractive and user-friendly interfaces.
- Presenting technical information simply.
- Integrating images, diagrams, and documentation.

Learning outcome

We learned how to transform technical information into a structured digital presentation for technical and non-technical users.

4. Learning Area 2 – Application Development

We explored application development and learned the difference between writing code and designing a user-oriented solution.

Key areas learned

- Understanding the application purpose before development.
- Planning user interface and interaction.

- Organizing application functions.
- Connecting functionality with the interface.
- Understanding usability.
- Testing and improving based on observed problems.

Learning outcome

Application development requires both technical implementation and user-centric thinking.

5. Learning Area 3 – 3D Modelling Using Autodesk Fusion 360

Fusion 360 introduced us to the mechanical side of product development.

Key areas learned

- Creating sketches and basic 3D shapes.
- Understanding dimensions and constraints.
- Converting sketches into components.
- Designing product enclosures.
- Considering component placement and wiring.
- Thinking about manufacturability and assembly.

Learning outcome

Mechanical design must accommodate actual components, wiring, accessibility, and intended operation.

6. Learning Area 4 – Custom PCB Design Using KiCad

KiCad introduced us to the transition from jumper-wire prototypes toward an integrated electronic system.

Key areas learned

- Understanding schematic design.
- Assigning footprints.
- Understanding PCB layers and copper zones.
- Using ground planes.
- Routing traces and placing components.
- Running Design Rule Checks.
- Identifying and correcting PCB errors.

Learning outcome

PCB design requires accuracy, planning, electrical awareness, and careful verification.

7. Learning Area 5 – Critical Thinking

Rather than immediately implementing the first solution, we learned to question the design and evaluate whether a particular approach was appropriate.

Questions used during evaluation

- What is the actual problem we are trying to solve?
- Is the proposed solution practical?

- Is every component necessary?
- Can the design be simplified?
- What could fail?
- How will the user interact with the system?
- Can the design be manufactured?
- Is the selected technology appropriate for the current phase?
- What improvements can be made in the next phase?

This approach helped us move from building something simply because it can be built toward building something because it solves a defined problem effectively.

8. Problem-Solving and Decision-Making

Throughout the learning process, different approaches to hardware integration, enclosure design, PCB implementation, and system presentation were evaluated before deciding what was appropriate for the current stage.

This taught us that a good solution is not necessarily the most complicated solution; it is the solution appropriate for the problem, requirements, and current stage of development.

The process improved our ability to compare alternatives, identify limitations, and make informed decisions through feedback, testing, and reflection.

9. Multidisciplinary Learning

A major outcome of the learning journey was understanding how different engineering disciplines connect. These areas are interconnected rather than isolated.

Learning area	Contribution
Website development	Digital presentation and documentation
Application development	User interaction and software interface
Fusion 360	Mechanical and enclosure design
KiCad	Electronic and PCB design
Embedded systems	Hardware control and automation
Critical thinking	Problem analysis and decision-making
Documentation	Professional communication

Learning flow: Idea → Problem analysis → Electronics → Embedded control → Mechanical design → Software/application → Website → Documentation → Prototype

10. Skills Developed

Technical skills

- Website development
- Application development

- 3D CAD modelling
- PCB schematic design
- PCB layout
- Hardware integration
- Embedded-system understanding
- Technical documentation

Engineering skills

- System-level thinking
- Component selection
- Design evaluation
- Prototyping
- Troubleshooting
- Design verification

Professional skills

- Critical thinking
- Problem-solving
- Decision-making
- Communication
- Technical presentation
- Team collaboration
- Iterative learning

11. Overall Learning Outcome

The most significant outcome was learning to look at an engineering problem from multiple dimensions. Developing a complete product involves software, hardware, mechanical design, user interface, documentation, and critical thinking.

This multidisciplinary approach improved our confidence in developing future projects and provided a stronger foundation for innovation and product development.

12. Conclusion

The AquaSense Phase 1 learning journey provided practical exposure to important technologies and engineering practices, including website development, application development, Fusion 360 modelling, KiCad PCB design, and critical thinking.

We learned that innovation involves understanding the problem, evaluating alternatives, designing systematically, testing assumptions, learning from mistakes, and continuously improving the solution. This foundation can be applied to future engineering projects, prototypes, and entrepreneurial ideas.